

Route 53 Failover Routing

Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service.

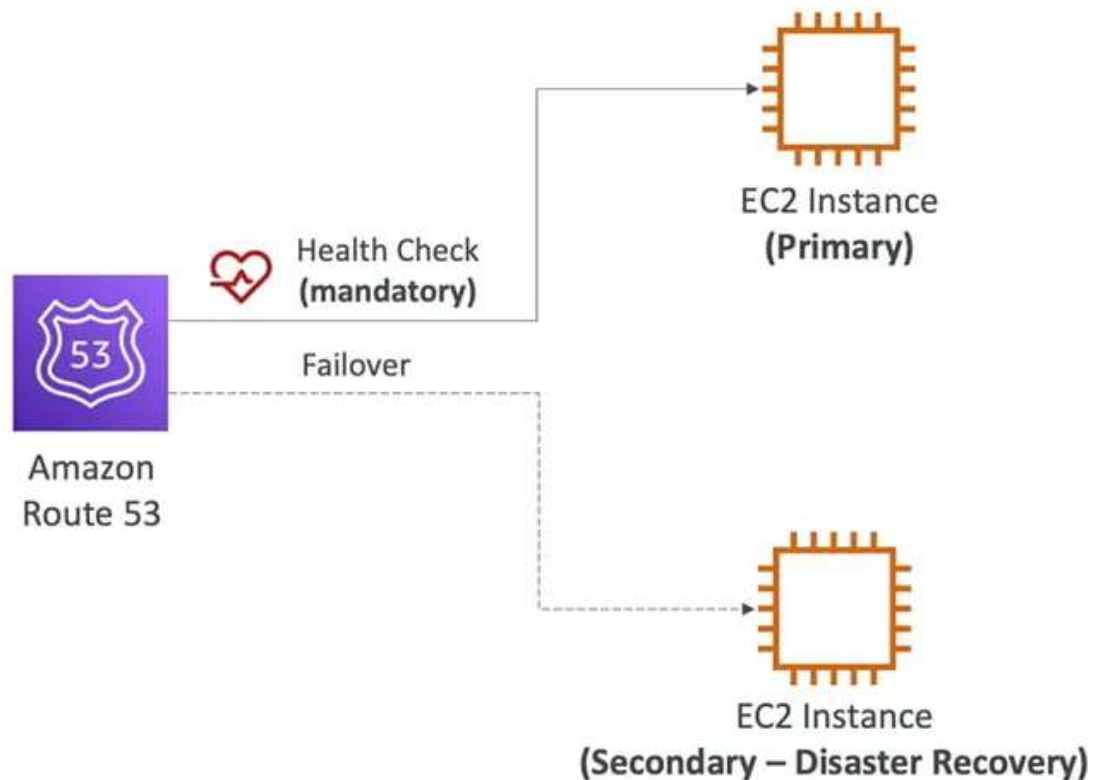
Route 53 can be used to perform three main functions:

1. **Domain registration:** if your website needs a name, such as name.com. Route 53 eases the registration of naming your website or web application
2. **DNS routing- Route internet traffic to the resources for your domain:** When a user opens a web browser and enters your domain name (name.com) or subdomain name (abc.name.com) in the address bar, Route 53 helps connect the browser to your website/web application.
3. **Health check:** Route 53 sends automated requests over the internet to a resource, such as a web server, to verify that it's reachable, available, and functional. In addition, you can setup SNS to receive notifications when a resource becomes unavailable and choose to route internet traffic away from unhealthy resources.

LAB OVERVIEW:

Configure your domain in a manner that if the website in the primary Availability Zone becomes unavailable, Amazon Route 53 will automatically fail over application traffic to the instance in the secondary Availability Zone.

(Basically, number 2 from above — DNS ROUTING)



User requests are normally sent to the IP address in Primary Instance. If Primary Instance is unavailable, requests are routed to the Secondary Instance based on the configuration in the Route 53 records. When Primary Instance becomes unavailable, a Route 53 health check alarm is invoked, and possibly an email alert is sent to the email address configured. (Be sure to have configured **two identical instances/websites** in different availability zone for best result)

OBJECTIVES:

1. Configure a Route 53 health check that sends emails via SNS when the health of an HTTP endpoint becomes unhealthy.
2. Configure failover routing in Route 53.

TO- DO 1: Configure Route 53 Health check:

The first step to configure failover is to create a health check for your primary website. In the AWS Management Console, from the **Services** menu, enter and choose Route53 to open the **Route 53 Management Console**.

1. In the left navigation pane, choose **Health checks**.
 2. Choose **Create health check**, and configure the following options. Leave the default values for all other fields.
 - **Name:** Enter 'Primary health check'
 - **What to monitor:** Choose **Endpoint**.
 - **Specify endpoint by:** Choose **IP address**.
 - **IP address:** Paste in the **Public IPv4 address** of **PrimaryInstance**.
 3. Expand **Advanced configuration**,
 - **Request interval:** Choose **Fast (10 seconds)**.
 - **Failure threshold:** Enter '2'
- N.b These options make your health check respond faster.
4. Choose **Next**.
 5. **Get notified when health check fails**, configure the following options:
 - **Create alarm:** Choose **Yes**.

- **Send notification to:** Choose **New SNS topic**
- **Topic name:** Enter 'primaryhealthcheck'
- **Recipient email address:** Enter an email address that you can access, confirm subscription
- **Create health check.**

N.B — **Route 53** now checks the health of your site by requesting the domain name you provided, verifying that it returns a successful response else it triggers a notification.

Create health check

Step 1: Configure health check

Step 2: Get notified when health check fails

Configure health check

Route 53 health checks let you track the health status of your resources, such as web servers or mail servers, and take action when an outage occurs.

Name Primary health check ⓘ

What to monitor ☒ Endpoint ⓘ

☐ Status of other health checks (calculated health check)

☐ State of CloudWatch alarm

Monitor an endpoint

Multiple Route 53 health checkers will try to establish a TCP connection with the following resource to determine whether it's healthy.

[Learn more](#)

Specify endpoint by ☒ IP address ☐ Domain name

Protocol HTTP ⓘ

IP address * 44.2.0.0000000 ⓘ

Host name www.example.com ⓘ

Port * 80 ⓘ

Path /images ⓘ

Advanced configuration

Create health check

Step 1: Configure health check

Step 2: Get notified when health check fails

Get notified when health check fails

If you want CloudWatch to send you an Amazon SNS notification, such as an email, when the status of the health check changes to unhealthy, create an alarm and specify where to send notifications.

Create alarm ☒ Yes ☐ No

CloudWatch sends you an Amazon SNS notification whenever the status of this health check is unhealthy for at least one minute. The alarm will be located in the **us-east-1** region.

Send notification to ☐ Existing SNS topic ☒ New SNS topic

Topic name * PrimaryHealthCheck

Recipient email addresses * you@me.com

Separate multiple addresses with a comma, a pipe character, or a space

* Required

Cancel

Previous

Create health check

Health checks console feedback collection

To help us improve the Health Check user experience, please take 5 minutes to complete this survey.

Create health check

Delete health check

Edit health check

Filter by keyword

<< < 1 to 1 of 1 health check > >>

Name	Status	Description	Alarms	ID
☐ Primary health check	15 minutes ago Healthy	http://44.200.000.000.000.000	1 of 1 in INSUFFICIE...	d#3f009-159b-49

Info Monitoring Alarms Tags Health checkers Latency

No health check selected.

No health check selected.

Fig. 2 Configuring health checks

TO-DO 2.1 : Configure Route 53 records

In the Route 53 console, in the left navigation pane, choose **Hosted zones**.

1. Choose **Create record**, and configure the following options:

- **Record name:** Enter 'failover'

- **Record type:** Choose **A — Routes traffic to an IPv4 address and some AWS resources.**
- **Value:** enter the IP address for **PrimaryInstance.**
- **TTL (seconds):** Enter “15”
- **Routing policy:** Choose **Failover.**
- **Failover record type:** Choose **Primary.**
- **Health check ID:** Choose **Primaryhealth check**
- **Record ID:** Enter “primaryfailover”
- Choose **Create records.**

T0-D0 2.2: Creating an A record for the secondary website

Create another record for the stand-by/secondary web server.

1. Choose **Create record**, and configure the following options
 - **Record name:** Enter “failover”
 - **Record type:** Choose **A — Routes traffic to an IPv4 address and some AWS resources.**
 - **Value:** enter the IP address for **SecondaryInstance.**
 - **TTL (seconds):** Enter “15”
 - **Routing policy:** Choose **Failover.**
 - **Failover record type:** Choose **Secondary.**
 - **Health check ID:** Leave this field empty.

- **Record ID:** Enter “SecondaryFailover”
- Choose **Create records**.

Quick create record [Switch to wizard](#)

▼ **Record 1** Delete

Record name [Info](#) .60 Record type [Info](#)

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

Enter multiple values on separate lines.

TTL (seconds) [Info](#) Routing policy [Info](#)

Recommended values: 60 to 172800 (two days)

Failover record type
Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.

Record ID [Info](#)

Health check ID [Info](#)

Primary health check
ID: dff9f009-159b-495b-bf0a-cee7dc93f177

Record name [Info](#) .60 Record type [Info](#)

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

Enter multiple values on separate lines.

TTL (seconds) [Info](#) Routing policy [Info](#)

Recommended values: 60 to 172800 (two days)

Failover record type
Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.

Record ID [Info](#)

Health check ID - optional [Info](#)

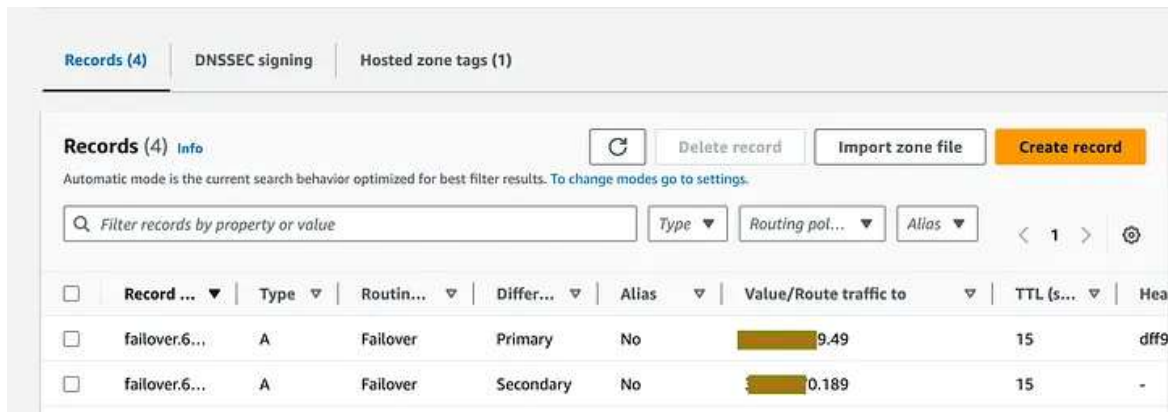


fig. 3 Configuring Route 53 records

T0-D0 3: Visit the DNS records in a browser to verify that Route 53 is pointing correctly to your primary website.

1. Select the check box for either one of the A records. A **Record details** panel appears that includes the **Record name**. Copy the **Record name** value of the A record and Open a new browser tab. Paste the A record name and launch.
2. Verify the failover functionality — verify that Route 53 correctly fails over to your secondary server if your primary server fails.

N.B: You can simulate a failure by manually stopping the **PrimaryInstance**.

When the primary instance/website stops functioning. The Route 53 health check that you configured notices that the application is not responding, and the record entries configured causes DNS traffic to fail over to the secondary EC2 instance.

3. Check your email, by now you should have received an email from AWS Notifications — SNS titled “ALARM: Primary-Health-awsroute53-xxxxxx” stating the cause of the alarm.